

[Aim: 100/100 in Maths]

आरंभ

# ARITHMETIC PROGRESSION

Lecture #4

#ATK<sup>2</sup>:-

(d)  $\rightarrow$  same.

$n^{\text{th}}$  term of AP:  $a_n = a + (n-1)d$

~~Q3~~  $\rightarrow$  कौनसी / कौसी / बिना  $\rightarrow$  'n' term

(\*) Q में AP नहीं बोली गई थी।

$\rightarrow$  सबसे पहला साध  $\rightarrow$  AP prove करोगे  
then formula लगाओगे

#LP: Which term of the arithmetic progression 5, 15, 25, ... will be 130 more than its 31<sup>st</sup> term? [CBSE 2006, 2017]

AP: 5, 15, 25 - - -

Let  $n^{\text{th}}$  term is 130 more than 31<sup>st</sup> term.

$$d = 15 - 5$$

$$d = 10$$

$$a_n = 130 + a_{31}$$

$$a + (n-1)d = 130 + a + 30d$$

$$(n-1)(10) = 130 + 30(10)$$

$$10n - 10 = 130 + 300$$

$$10n = 430 + 10$$

$$10n = 440 \Rightarrow n = 44$$

$\therefore 44^{\text{th}}$  term will be 130 more than 31<sup>st</sup> term.

#LP: Which term of the A.P. 121, 117, 113, ... is its first negative term?

Let  $n^{\text{th}}$  term is first -ve term.

$$d = 117 - 121$$

$$d = -4$$

$$a_n < 0$$

$$a + (n-1)d < 0$$

$$121 + (n-1)(-4) < 0$$

$$\Rightarrow 121 - 4n + 4 < 0$$

$$\Rightarrow 125 - 4n < 0$$

$$\Rightarrow 125 < 4n$$

$$\Rightarrow \frac{125}{4} < n$$

$$\Rightarrow n > \frac{125}{4}$$

$$n > 31.25$$

$\therefore$  32nd term is first -ve term.

$n \rightarrow$  term?  
Natural no.



#LP: Is -150 a term of the A.P. 11, 8, 5, 2, ...?

AP

11, 8, 5, 2, - - - - -

-150

??

Let  $n^{\text{th}}$  term

$$d = 8 - 11$$

$$d = -3$$

Let -150 is  $n^{\text{th}}$  term

$$\therefore a_n = -150$$

$$a + (n-1)d = -150$$

$$11 + (n-1)(-3) = -150$$

$$(n-1)(-3) = -150 - 11$$

$$(n-1)(-3) = +161$$

$$n-1 = \frac{161}{3}$$

$$n = \frac{161}{3} + 1$$

$$n = \frac{161+3}{3}$$

$$n = \frac{164}{3} \text{ Sy. 6.}$$

$n$  is always natural no.  
or -150 is not a term of this AP.

#LP: Is 184 a term of the sequence 3, 7, 11, ....?

check AP or not?? Yes

∴ AP. 3, 7, 11, ...

Let 184 is  $n^{\text{th}}$  term

$$\therefore a_n = 184$$

$$a + (n-1)d = 184$$

$$3 + (n-1)(4) = 184$$

$$(n-1)(4) = 184 - 3$$

$$(n-1)(4) = 181$$

$$n-1 = \frac{181}{4}$$

$$n = \frac{181}{4} + 1$$

$$n = \frac{181+4}{4}$$

$$n = \frac{185}{4} = 46.25$$

Since  $n$  is not a Natural No.  $\therefore 184$  is not a term of this sequence

$$\begin{matrix} d=7-3 \\ d=4 \end{matrix}$$

**#LP: Determine the 10<sup>th</sup> term from the end of the A.P. 4, 9, 14, ..., 254.**

AP: 4, 9, 14, ..., 254.

← 10<sup>th</sup>

Reverse AP: 254, ..., 14, 9, 4

→  $a_{10} = a + 9d$

$$= 254 + 9(9 - 14)$$

$$= 254 + 9(-5)$$

$$= 254 - 45$$

$$a_{10} = 209$$

∴ 10<sup>th</sup> term from last  
of the given AP: 209

#LP: Find the 8<sup>th</sup> term from the end of the A.P. 7, 10, 13, ..., 184  
[ CBSE 2005 ]

Reverse AP: 184, - - -, 13, 10, 7  
8<sup>th</sup>

$$\begin{aligned}a_8 &= a + 7d \\&= 184 + 7(10 - 13) \\&= 184 + 7(-3)\end{aligned}$$

$$a_8 = 184 - 21$$

$$\boxed{a_8 = 163} \quad \swarrow$$



~~Q. 1~~ Q. 1  $n^{\text{th}}$  term given  $\rightarrow n = 1, 2, 3, \dots$  put  $n=1$   
#LP: The  $n^{\text{th}}$  term of an A.P. is  $6n + 2$ . Find the common difference.

[CBSE 2006]

$$a_n = 6n + 2$$

$$n=1 \rightarrow a_1 = 6(1) + 2$$

$$a_1 = 8 = a$$

$$n=2 \rightarrow a_2 = 6(2) + 2$$

$$a_2 = 14$$

(ii) Find  $30^{\text{th}}$  term ??

$$a_{30} = a + 29d$$

$$a_{30} = 8 + 29(6)$$
$$= 8 + 174$$

$$a_{30} = 182$$

$$d = ??$$

$$a_2 - a_1 = ??$$

$$\Rightarrow d = a_2 - a_1$$

$$= 14 - 8$$

$$d = 6$$

#LP: The general term of a sequence is given by  $a_n = +4n + 15$ . Is the sequence an A.P.? If so, find its 15th term and the common difference.

$$a_n = 4n + 15$$

$$n=1 \longrightarrow a_1 = 4(1) + 15$$

$$a_1 = 19 \checkmark = a$$

$$n=2 \longrightarrow a_2 = 4(2) + 15$$

$$a_2 = 23 \checkmark$$

$$n=3 \longrightarrow a_3 = 4(3) + 15 \Rightarrow 27 \checkmark$$

$$n=4 \longrightarrow a_4 = 4(4) + 15 \Rightarrow 31 \checkmark$$

(i) AP?

$$d_1 = a_2 - a_1 = 23 - 19 \Rightarrow 4$$

$$d_2 = a_3 - a_2 = 27 - 23 \Rightarrow 4$$

$$d_3 = a_4 - a_3 = 31 - 27 \Rightarrow 4$$

Since all  $d$ 's are equal.

Yes it's AP

$$(i) a_{15} = a + 14d$$

$$= 19 + 14(4)$$

$$a_{15} = 19 + 56$$

$$a_{15} = 75$$

$$(ii) d = 4$$

#123B

Middle term

$$\text{Pos} \Rightarrow \left(\frac{n+1}{2}\right)^{\text{th}} = \left(\frac{11+1}{2}\right)^{\text{th}} \\ = \left(\frac{12}{2}\right)^{\text{th}} \rightarrow \textcircled{6^{\text{th}}}$$

3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33  
 $a_1, a_2, a_3, a_4, a_5, \text{6}^{\text{th}}, a_7, a_8, a_9, a_{10}, a_{11}$   $\rightarrow$  11 terms

$n=11$

① Position of M.T.  $\Rightarrow$   $6^{\text{th}}$  term

② Value of m.T.  $\Rightarrow$   $a_6$   
( $a+5d$ )

Position of MT

odd  $\rightarrow$  Pos. of MT =  $\left(\frac{n+1}{2}\right)^{\text{th}}$

$n$  even  $\rightarrow$  Pos. of MT  $\Rightarrow$   $\left(\frac{n}{2}\right)^{\text{th}}$  &  $\left(\frac{n}{2} + 1\right)^{\text{th}}$



#LP: Find the middle term of the A.P. 6, 13, 20, ..., 216. [ CBSE 2015 ]

AP: 6, 13, 20, - - - 216  
 $a_1, a_2, a_3$   
let  $n$  terms.

$$\therefore a_n = 216$$

$$a + (n-1)d = 216$$

$$6 + (n-1)(7) = 216$$

$$(n-1)(7) = 210$$

$$n-1 = 30$$

$$n = 31$$

$$n = 31 \leftarrow \text{odd}$$

(i) position of MT

$$\left(\frac{n+1}{2}\right)^{\text{th}} \Rightarrow \left(\frac{31+1}{2}\right)^{\text{th}} \\ = \left(\frac{32}{2}\right)^{\text{th}} = 16^{\text{th}}$$

$$a_{16} = a + 15d \\ = 6 + 15(7) \\ = 6 + 105 \\ \boxed{a_{16} = 111}$$

- ① n find
- ① position of MT  
 $\rightarrow n = ?$
- ② Value of MT



1   2   3

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0.6

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31

16<sup>th</sup>  
↑  
m7

#LP: Find the middle term(s) of the A.P. 7, 13, 19, ..., 241.

AP: 7, 13, 19, ..., 241

let  $n$  terms are there.

$\therefore a_n = 241$

$$a + (n-1)d = 241$$

$$7 + (n-1)(6) = 241$$

$$(n-1)(6) = 241 - 7$$

$$(n-1)(6) = 234 \quad 39$$

$$n-1 = 39$$

$$\boxed{n=40}$$

(i) Pos. of MT  $\boxed{n=40}$  — even

$$\left(\frac{n}{2}\right)^{\text{th}} \quad \left(\frac{n}{2} + 1\right)^{\text{th}}$$

$$\rightarrow \left(\frac{40}{2}\right)^{\text{th}} \quad , \quad \left(\frac{40}{2} + 1\right)^{\text{th}}$$

$$\rightarrow \underline{\underline{20^{\text{th}} \text{ \& } 21^{\text{th}}}}$$

①  $n$  find.

① Position of MT

② Value of MT



$$\begin{aligned} a_{20} &= a + 19d \\ &= 7 + 19(6) \\ &= 7 + 114 \end{aligned}$$

$$\boxed{a_{20} = 121}$$

$$\begin{aligned} a_{21} &= a + 20d \\ &= 7 + 20(6) \end{aligned}$$

$$\boxed{a_{21} = 127}$$

$\therefore$  MTs are 121 & 127

✓

HW

**#LP: Consider the A.P. 2 , 5 , 8 , 11 , ... , 302. Show that twice of the middle term of the above A.P. is equal of the sum of its first and last term.**





AP  $\rightarrow$  module

**THANK YOU COODIESS !!**

